Identification of Non-Vehicle and Vehicle Density at Intersections Based on GPS Data

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***Abstract*— This study aims to identify the density of vehicles and non-vehicles at the Zero Point Yogyakarta intersection based on GPS data. The data in this study comes from one of the telecommunication operator companies in Indonesia with active MPD data criteria. The data was collected from telecommunication service users who consciously activated their location services, during the period from November 2021 to January 2022. Zero-Point Intersection in Yogyakarta was chosen because of its high density, historical value, and location in the center of Yogyakarta and as a tourist destination. In this study, GPS data was processed, using a rules-based approach to categorize users into vehicle and non-vehicle groups based on speed, distance, mobility time, and duration of time at the intersection. The results of the analysis show that the highest density occurred in November 2021, with a higher number of vehicles and non-vehicles during the day than at night. The decrease in density was seen in December 2021 and continued until January 2022, due to movement restrictions during the COVID-19 pandemic. The data shows that the number of non-vehicle users is higher than that of vehicles, especially during the day. The rule-based method used in this study succeeded in distinguishing between vehicles and non-vehicles, as well as providing an understanding of mobility patterns at the Zero Point intersection in Yogyakarta. The study shows that the use of GPS technology and rule-based data analysis can enhance ongoing research on traffic management efficiency in congested urban areas.**

Keywords—GPS Data, Vehicle, Zero-point intersection, non-vehicle, Yogyakarta

I. Introduction

In the era of rapidly developing information technology, GPS (Global Positioning System) data has become very important in various fields, including city traffic management [1]. Using GPS data to identify vehicle and non-vehicle behavior provides an excellent opportunity to optimize the transportation system, especially in strategic and historical locations such as the Titik Nol Yogyakarta (Zero-Point intersection in Yogyakarta). This intersection, located in the heart of Yogyakarta, connects various social, economic, and cultural activities [2].

Zero-Point intersection in Yogyakarta serves not only as a connecting road but also as a meeting area for various elements of society, both vehicle users and pedestrians. High traffic density often challenges driving efficiency and safety, while pedestrians' need for a safe and comfortable space must also be considered [3] [4]. In this context, GPS data-based vehicle and non-vehicle behavior identification becomes highly relevant.

This study collected and analyzed GPS data derived from Indonesian telecommunication cellular location data to identify and understand vehicle and pedestrian movement patterns at Zero-Point intersection in Yogyakarta [5], thus filtering mobility based on behavior in mobility. Zero-Point intersection in Yogyakarta was chosen because it is one of the intersections of dense activities.

This work contributes to identifying vehicle and non-vehicle users based on actual GPS data in the area of interest, namely Yogyakarta Zero Point or Zero-point intersection in Yogyakarta. The GPS data used allowed researchers to distinguish between users who were driving and those who weren't, providing deeper insights into movement patterns in the area. Yogyakarta's Zero Point is an important central location, so this analysis can help in understanding the pattern of community mobility in the city center.

The paper is organized as follows. Section II discusses previous works related to the topic of this research. In section III, we provide the research method. In section IV, we discuss the results of the work. Finally, we conclude our work in Section V.

II. Related Works

The methodological approach employed for this study uses stratified random sampling technique in defining the study area based on productivity categories (high, medium selective and low). The samples in the study were then determined by using the Slovin method. The Comprehensive interview with salt farmers Provided inputs on factors affecting productivity During these interviews, the insurance companies document the outcomes in a causality diagram such as butterflies diagram, Fishbone diagram or Ishikawa diagram to point outcomes of causalities from people to process, to engine, and Human Resources [6].

At last, the model used in the research is Rule-Based System using forward chaining method. This method having used for expert systems development in research. Rule-based systems are designed from specialist knowledge in order to be able to receive data and make conclusions or guesses, developed [7].

Annur et al. Performed research on Rule-Based Expert System based system for determination of vowel from block of Arabic Text. Those researchers formed a model to identify

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appropriate vowels according to Arabic rhetoric by aligning multiple instances. This leads them to conclude that the system developed helps in understanding and applying harakat in Arabic text. Nonetheless, it must explicitly quantify the numerical or statistical strength of the model [8].

It introduces a list of arousal values (related to the energy experienced by the listener) and valence (related to the listener's pleasure) extracted from the audio and lyrics of the song and compiles a rules-based emotion detection methodology for songs. The primary aim was to find Thayer's mood labels along two other axes of arousal and valence dimensions. In other MIR papers, you will find that identifying music emotions is done through the analysis of audio alone [9], or a deep mix of audio, lyrics, and other resources, only audio, or other types of fusion.

In this study, Niharika et al. The image analysis method on fire detection using 20 samples of images with and without fire characteristics, demonstrated accuracy and precision percentages much higher than those achieved by the Vibe method. Rule based method accuracy is 99.19% and for the Vibe method, it is 86.24%. Vibe performs 10% better than rule-based methods at 60.19% precision [10].

Research conducted by Rakhman et al. Low sampling rate is the process of collecting GPS data at sparse or irregular time intervals that result in location points less frequent than more frequent location points or not showing every point of movement. The first role of low sampling rates is to lower the volume of data that must be stored, transmitted, and processed that can be stored, saving costs and resources [11].

Then the research conducted by Serang said that the analysis of land use and land cover changes in the Biyonga Sub-watershed, Gorontalo Regency showed that the use of Google Earth and KML (Keyhole Markup Language) for remote sensing data is a very easy method in visualizing and analyzing geographic data. KML can be used to store and share geographic data between different platforms – KML was developed for use with Google Earth. KML is capable of transmitting different types of data to end users through dynamic map displays, which helps users understand information with geographic analogues and allows users to make appropriate decisions. This integration has wide applications for education, research, and natural resource management [12].

Yogyakarta City is a city of culture, students, and tourism so many of these immigrants are both from outside the city and from abroad documenting Fransisca. This increase in the number of people is also accompanied by more vehicle movement, so managing the flow of traffic and the capacity of the road to avoid congestion is very important. In this study, the performance of the Ngabean signaling intersection had been assessed in order to optimize it. A field survey was carried out on March 5, 2020 by three seasons (morning, afternoon and evening) in collecting road geometric data, vehicle volume, speed, and side obstacles. This intersection has "one-way" and "two-way" track criteria with a waiting time of 153.81 seconds and a saturation degree of 1.55. Three other green-timings and an additional road widening improvements were proposed [13].

Then in this study Damayanto et al. discussed the evaluation of the level of pedestrian or non-driving service on Jalan Braga, Bandung using the Greenshields and Greenberg methods. The researcher analogized pedestrians with characteristics like vehicle traffic, which include flow, speed, and density. With pedestrian intersections having criteria for the presence of seats, light poles and urban plants, the average pedestrian speed is 0.36 km/h and the diameter of each pedestrian is 1.54 meters [14].

III. Methodology

Using radio signals encoded with digital data, satellites in a global positioning system (GPS) may pinpoint an object's precise location on Earth. Full functioning of the Global Positioning System (GPS) with 24 satellites occurred in 1994, making it a worldwide navigation system that gives data on time, location, speed, and direction [15]. Thanks to the coordinated efforts of satellites, ground stations, and receivers, Global Positioning System (GPS) provides precise location and timing information. By triangulating signals from at least four satellites, the receiver may reliably derive its position in terms of latitude, longitude, and, in certain cases, altitude. By periodically recording coordinates and time, GPS enables movement tracking and mobility pattern analysis; however, this GPS has a low sample rate [16].

A. Dataset

This study analyzed individual movement data collected from November 2021 to January 2022. This data was obtained from one of the telecommunication operator companies in Indonesia, the data collected information from active MPD (Mobile Positioning Data) users who consciously activated their location services. This data includes the user's GPS coordinates, with a low sampling rate. The scope of this data is the location of users around the Zero Point intersection in Yogyakarta. This data was collected for the purpose of analyzing Vehicle and Non-vehicle movement patterns. Data collection is carried out in connection with user privacy and compliance with applicable regulations.

TABLE I. Data Description

Information	Description
maid	Encrypted unique user ID
latitude	The approximate longitude coordinates of the device's location.
longitude	Approximate latitude coordinates of the location of the device.
timestamp	The time (in UNIX form) when the data was sampled.

Table I displays GPS data, where "maid" houses the Unix code for the GPS user data, "latitude" stands for the user's longitude coordinates, "Longitude" for the user's latitude coordinates, and on the time "timestamp" (in the form of UNIX "Uniplexed Information and Computing Service") when the data was sampled. In order to determine the current time, the Unix timestamp adds up all the seconds that have passed since the Unix Era began on January 1, 1970, at 00:00:00 UTC (without including leap seconds, though), as stated by Rakhman. Put simply, it is a time representation standard that is extensively utilized across many protocols and operating systems on the internet. The timestamp must be adjusted before the analysis can be carried out.

The location of our research is the intersection of Zero-Point in Yogyakarta. This intersection was chosen because of its high density, historical value, location in the center of Yogyakarta, and as a tourist destination. This intersection has

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an average waiting time of 153.81 seconds and a "one-way" intersection track on the north side and a "two-way" on the other side.

Data is collected at specific times and locations. In our study, activities were divided into two time criteria: day and night. Daytime activities take place from 06.00 to 17.59 WIB, while night activities are from 18.00 to 05.59 WIB [17][18].

B. Intersection Data

Intersection data is obtained by providing the intersection distance taken on Google Earth in KML format. Fig. 1, is a visualization of intersections in Google Earth.

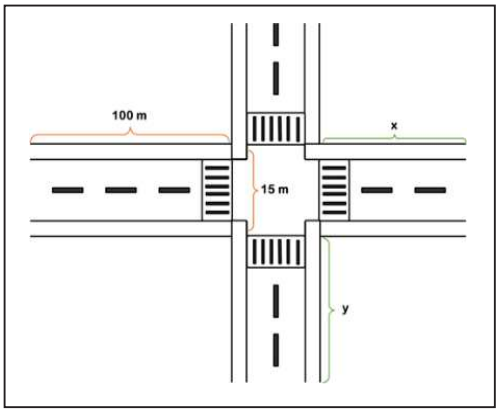

Fig. 1. Distance visualization and x, y axis.

Fig. 2. Google earth intersection.

We checked the Yogyakarta zero-point intersection, which has a length criterion of 100 meters or more, because according to Rosa [19] the intersection has a minimum length of 100 or more is a good intersection and can be monitored by CCTV (Closed Circuit Television), and for the width of the intersection we make it 15 meters because the width of the intersection track has a distance of 11 meters and the side of our intersection takes a distance of 2 meters each. In the orange braces symbol is the length of the intersection, then the green braces are the x and y axes, as shown in Fig. 1. Then Fig. 2, is a view Zero-Point intersection in the special region of Yogyakarta through Google earth. It was chosen because it has a waiting time duration of 153.81 seconds and has intersection criteria on the road, namely one and two directions on each road.

C. Research Flow

In this research flow, we describe the process from raw GPS data to produce a visualization of the comparison analysis of vehicle and non-vehicle users, as exemplified in Fig. 3.

In Fig. 3, the research model consists of several stages:

1) Raw GPS to represent GPS data before processing. Data was taken from November 2021 to January 2022 in Yogyakarta province.

2) Pre-processing of GPS data before the rule-based stage includes several important steps, such as using the Spatial Join technique to combine location data with the user's GPS data, performing Data Cleaning and Filtering to eliminate noisy data, ensuring each GPS user has a minimum number of data points with the Min Point Filter, as well as applying the Haversine technique to calculate the distance between GPS trajectory points and analyze the user's movement patterns.

3) Rule-based to group GPS data into vehicle and non-vehicle.

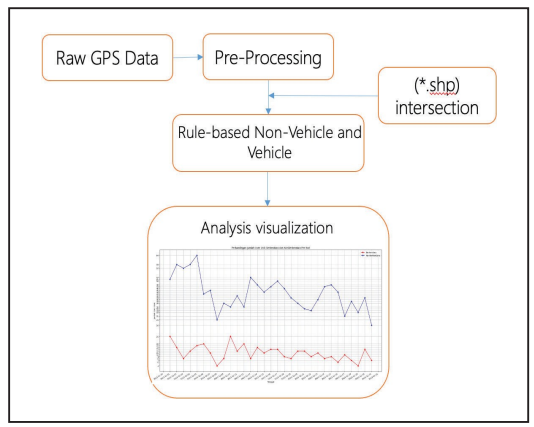

Fig. 3. Data comparison visualization.

D. Rule-based Vehicle and Non-Vehicle

We classify GPS data from vehicles and non-vehicles according to predetermined criteria using rule-based methods. We aim to facilitate analysis and decision-making based on the specific behavior or characteristics of each vehicle and non-vehicle group. Fig. 4, displays this flow that is based on rules.

In the rule-based flow, we show that GPS users at intersections are categorized as non-vehicle if they have speeds between 0.36 to 5 km/h, a distance of 11 meters, and a waiting time of 1 minute or more than 2.56 minutes. If the user does not meet these criteria, we will categorize them as vehicles.

IV. Results and Discussion

The study yielded several key results, which were given as follows:

A. Rule-based Vehicle and Non-Vehicle

In this study, we applied a rules-based process to group data into vehicles and non-vehicles. This process operates by identifying data according to predefined rules, as in Fig. 5.

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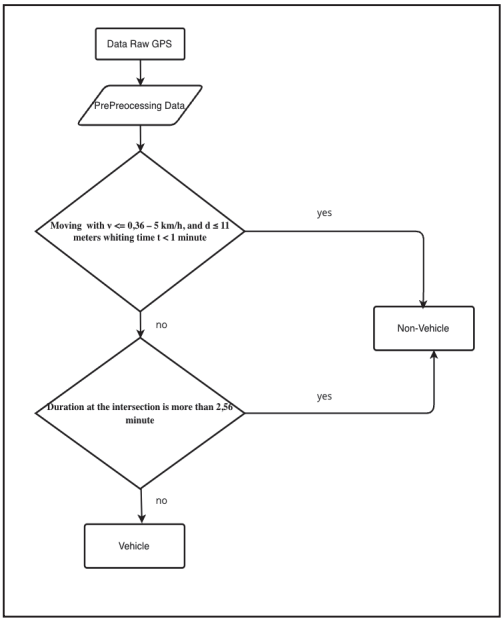

Fig. 4. Rule-based flow.

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• Identification of Non-Vehicle and Vehicle Density at Intersections Based on GPS Data
Input :
    - A group of GPS users at the Zero-Point intersection .
Output :
    - A group of GPS users at the Zero-Point intersection in Yogyakarta who were identified as driving an
non-driving
---------------------------------------------------------------------------------------------------
IF moving with v between 0.36 and 5 km/h, distance obtained d ≤ 11 meters in t < 1 minute OR IF duration
intersection > 2.56 minutes:
  return "Non-Vehicle"
else:
  return "Vehicle"
```

Fig. 5. Pseudocode non-vehicle and vehicle.

B. *Analysis of vehicles and non-vehicles*

Based on the results of our research analysis, we identified vehicles and non-vehicles using GPS data at Zero-Point intersection during day and night activity hours, daytime activities from 06.00 to 17.30 WIB, then night activities were carried out at 18.00 to 05.59 WIB. starting from November 2021 to January 2022. The results can be seen in Table II.

TABLE II. VEHICLE AND NON-VEHICLE DATA COMPARISON

GPS DATA						
Class	Nov21		Des21		Jan22	
	Day	Night	Day	Night	Day	Night
vehicle	1789	1062	1588	1064	875	499
non-vehicle	2016	1472	1390	1022	880	678
total	5409		4405		2520	

Table II is the result of a rule-based process that compares Vehicle and Non-vehicle GPS data users from November 2021 to January 2022 at the Zero-Point intersection in Yogyakarta. The table shows that the highest density occurred in November 2021, both during the day and at night, with 1789 vehicles and 2016 non-vehicles during the day, as well as 1062 vehicles and 1472 non-vehicles at night. In December 2021, the number of vehicles and non-vehicles decreased to 1588 and 1390 during the day, and 1064 and 1022 at night. This downward trend continued until January 2022, with the number of vehicles and non-vehicles being 875 and 880 during the day, and 499 and 678 at night. This decrease in density can be seen more clearly in Fig. 6 to 10.

Fig. 6 shows an increase in the number of mobile GPS users around the Zero Point Intersection area in Yogyakarta in early November 2021, more density occurs during daytime activities and is dominated by Non-vehicle, followed by daytime vehicle and both have decreased in night activities. This congestion occurred according to the Yogyakarta Special Region Tourism Office [20] scheduled several important events, including the Jogja Agricultural and Investment Culture Carnival, which led to an increase in density from November 1 to 10, 2021. After the event, our analysis showed a decrease in the density of GPS users at the intersection.

Fig. 7, Non-vehicle activities during the day increased in early December starting from December 1 to 5, 2021 and then decreased afterwards, in vehicle during daytime activities it was seen that the increase for 8 days occurred from December 13 to 20, 2021 but night vehicle activities increased earlier 4 days earlier than vehicle daytime activities, Then for non-vehicle night activities, there was a decrease at the end of December 2021 but an increase on December 31, 2021 during holidays but not significant.

Fig. 8, shows data for January 2021. This graph shows that although there is a slight fluctuation, the general trend shows a decrease in the number of GPS users for both vehicles and non-vehicles, day and night. The data shows for January 2022. The downward trend in GPS user density continued into this month, with the number of users stabilizing at lower numbers compared to previous months.

The data shows that the density of GPS users is declining, although December and January are usually the year-end and New Year's holiday months. The COVID-19 pandemic, travel restrictions, and changing holiday celebration habits led to this reduction [21].

V. CONCLUSION

The purpose of this study is to identify the density of vehicles and non-vehicles at the Zero-Point intersection in Yogyakarta using GPS data. This research focuses on how to distinguish active MPD data in the form of GPS data users, at intersections that have criteria such as various activities, density, specific waiting times, activities, and culture.

This study utilizes GPS data obtained from one of the telecommunication operator companies in Indonesia with the criteria of active MPD data, users who activate their location consciously. Data was taken between November 2021 and January 2022. The data is processed through a rules-based approach to categorize them into vehicle and non-vehicle groups, taking into account factors such as speed, distance, length, and waiting time at intersections. The procedure includes several phases, which include preprocessing GPS data, combining data, and analyzing movement patterns.

Our data shows that the peak of density occurred in November 2021 due to several events organized by the Yogyakarta Tourism Office, which took place both during the day and at night. Density decreased between December 2021 and January 2022, mainly due to the COVID-19 epidemic that restricted population movement. The data shows that the number of Non-vehicle vehicles exceeded vehicles in November 2021, especially during the day, but vehicle increased in December 2021 and January decreased in both. The study successfully used a rules-based strategy to classify GPS data and gain a detailed understanding of mobility patterns at Zero-Point intersections in Yogyakarta.

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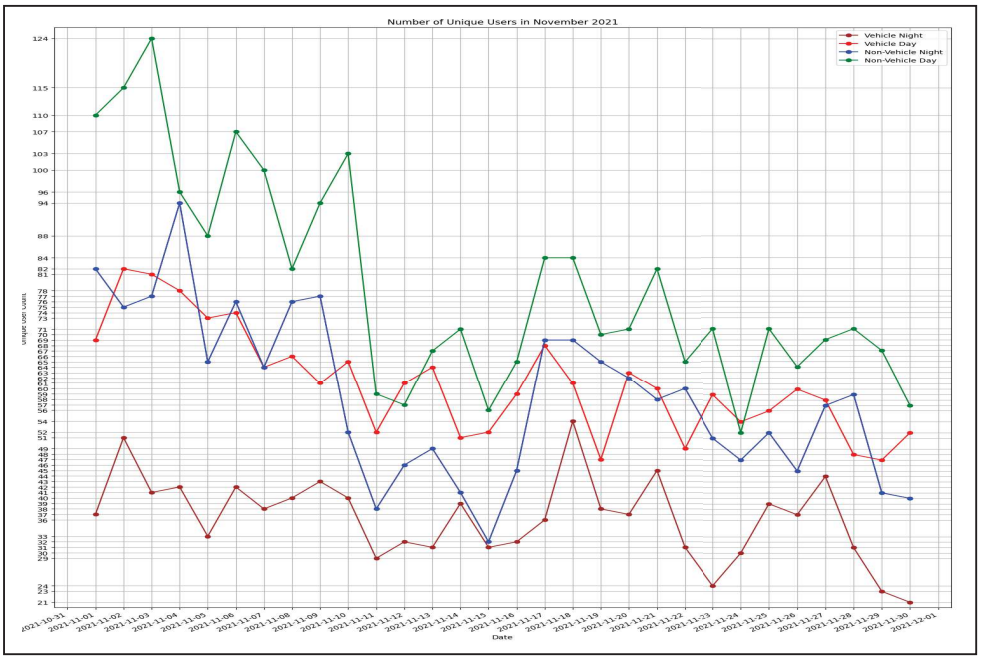

Fig. 6. November 2021.

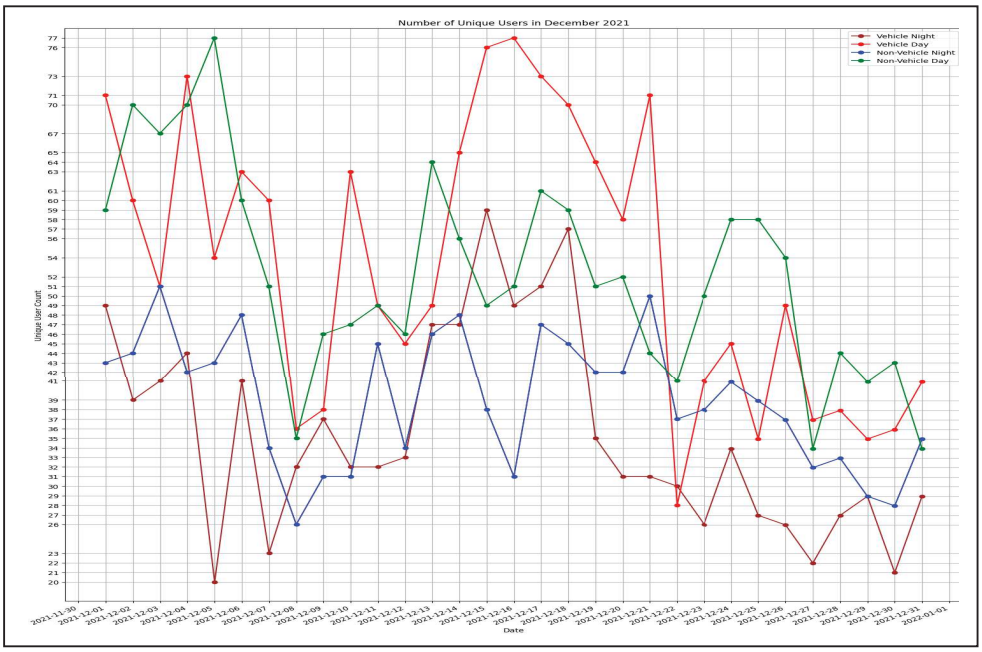

Fig. 7. December 2021.

This study effectively demonstrates the feasibility of utilizing GPS data to identify and assess vehicle and non-vehicle density at intersections that have congested criteria in urban areas. Rule-based procedures produce precise results and have the ability to group movement data. In summary, the study shows that the use of GPS technology and rule-based data analysis can enhance ongoing research, for further investigation of the application of comparable technologies in various fields and for broader traffic management purposes.

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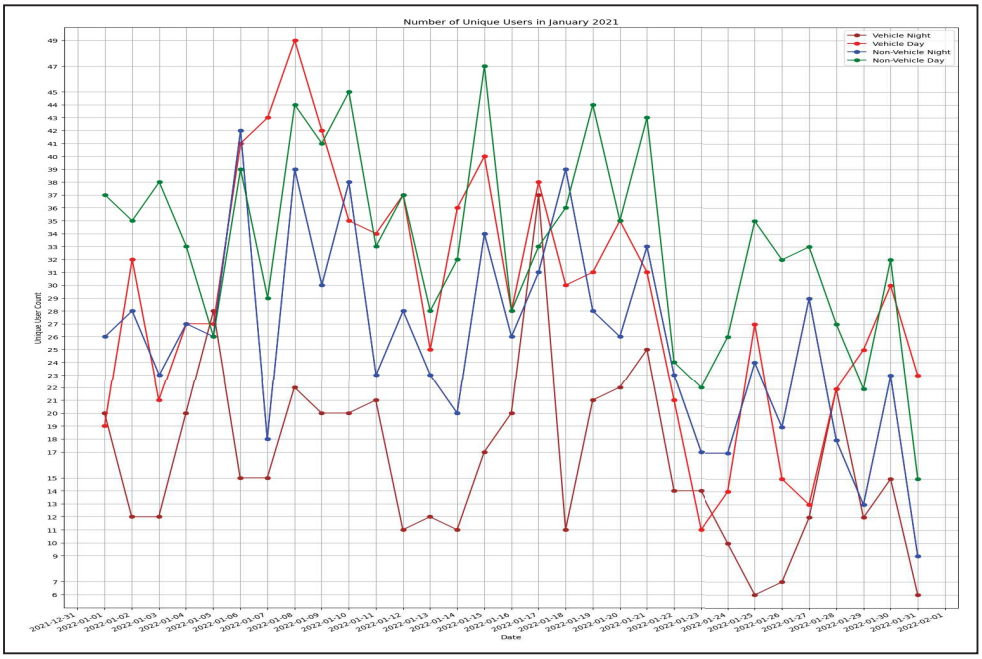

Fig. 8. January 2022.

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